

librispeech_validated_ensemble

scalars_validated_ensemble_librispeech

key	value
skipped_empty_gt	0
step2_stats.n_models	7
step2_stats.spearman	1
step2_stats.kendall_b	1
step2_stats.spearman_exact_p_onetailed	0.0002
step2_stats.kendall_exact_p_onetailed	0.0002
step2_stats.p_method	exact
step2_stats.n_permutations	5040
step2_stats.min_attainable_p	0.0002
step2_stats.pearson	0.94
step2_stats.pearson_p	0.0016
step2_stats.mean_wer_compression	0.5494
step3_determinism.identical_outputs	2611
step3_determinism.n	2611
step3_determinism.deterministic	True
step4_resolution_true.reference	true
step4_resolution_true.n_boot	10000
step4_resolution_true.n_samples	2611
step4_resolution_true.min_resolved_gap_wer	0.1414
step4_resolution_true.max_unresolved_gap_wer	0.0325
step4_resolution_true.n_resolved	20
step4_resolution_true.n_pairs	21
step4_resolution_pseudo.reference	pseudo
step4_resolution_pseudo.n_boot	10000
step4_resolution_pseudo.n_samples	2611
step4_resolution_pseudo.min_resolved_gap_wer	0.1606
step4_resolution_pseudo.max_unresolved_gap_wer	0.0949
step4_resolution_pseudo.n_resolved	17
step4_resolution_pseudo.n_pairs	21
step8_permutation_summary.canonical_order	seamless_m4t+whisper_v3+parakeet+sb+cohere
step8_permutation_summary.n_orders	6
step8_permutation_summary.n_transcripts	2611
step8_permutation_summary.total_differing	0
step8_permutation_summary.invariant	True

step0_cross_wer_validated_ensemble_librispeech

	seamless_m4t	whisper_v3	canary	conformer	parakeet	sb	cohere
seamless_m4t	0	2.84	2.21	2.38	2.14	2.56	2.14
whisper_v3	2.84	0	2.35	2.6	2.22	2.81	2.16
canary	2.21	2.35	0	1.9	1.27	2.16	1.54
conformer	2.38	2.6	1.9	0	1.6	1.89	1.6
parakeet	2.14	2.22	1.27	1.6	0	1.88	1.29
sb	2.56	2.81	2.16	1.89	1.88	0	1.74
cohere	2.14	2.16	1.54	1.6	1.29	1.74	0

step0_err_corr_validated_ensemble_librispeech

	seamless_m4t	whisper_v3	canary	conformer	parakeet	sb	cohere
seamless_m4t	1	0.5964	0.6161	0.6072	0.6198	0.5441	0.54
whisper_v3	0.5964	1	0.6137	0.5985	0.6236	0.5265	0.5666
canary	0.6161	0.6137	1	0.6336	0.7444	0.5804	0.5842
conformer	0.6072	0.5985	0.6336	1	0.6947	0.6515	0.61
parakeet	0.6198	0.6236	0.7444	0.6947	1	0.6234	0.6282
sb	0.5441	0.5265	0.5804	0.6515	0.6234	1	0.5721
cohere	0.54	0.5666	0.5842	0.61	0.6282	0.5721	1

step0_flags_validated_ensemble_librispeech

pair	cross_wer	err_corr	note
canary parakeet	1.27	0.7444	unusually similar hypotheses - suspect shared lab/data/teacher
canary cohere	1.54	0.5842	unusually similar hypotheses - suspect shared lab/data/teacher
conformer parakeet	1.6	0.6947	unusually similar hypotheses - suspect shared lab/data/teacher
conformer cohere	1.6	0.61	unusually similar hypotheses - suspect shared lab/data/teacher
parakeet cohere	1.29	0.6282	unusually similar hypotheses - suspect shared lab/data/teacher

step1_bias_validated_ensemble_librispeech

model	role	n_samples	spearman_full	spearman_loo	kendall_full	kendall_loo	delta_spearman	delta_kendall
cohere	voter	2611	0.4029	0.4275	0.3858	0.4076	0.0246	0.0218
parakeet	voter	2611	0.5876	0.6003	0.5627	0.5739	0.0127	0.0112
whisper_v3	voter	2611	0.6428	0.6729	0.6134	0.6433	0.0301	0.0299
sb	voter	2611	0.6646	0.6828	0.6352	0.653	0.0182	0.0178
canary	ranked_only(control)	2611	0.6737	0.6737	0.6438	0.6438	0	0
conformer	ranked_only(control)	2611	0.7172	0.7172	0.6849	0.6849	0	0

model	role	n_samples	spearman_full	spearman_loo	kendall_full	kendall_loo	delta_spearman	delta_kendall
seamless_m4t	voter	2611	0.7518	0.7772	0.7175	0.744	0.0254	0.0265

step2_ranking_validated_ensemble_librispeech

model	role	n_samples	wer_true	cer_true	wer_pseudo_loo	cer_pseudo_loo	rank_true	rank_pseudo	wer_compression
cohere	voter	2611	1.2245	0.3722	1.0833	0.3562	1	1	0.1412
parakeet	voter	2611	1.6677	0.4946	1.1573	0.3496	2	2	0.5103
canary	ranked_only	2611	1.8912	0.6029	1.2491	0.4156	3	3	0.642
conformer	ranked_only	2611	2.0325	0.5964	1.3179	0.3821	4	4	0.7146
sb	voter	2611	2.2063	0.6697	1.6315	0.4992	5	5	0.5749
seamless_m4t	voter	2611	2.5693	0.9011	1.9065	0.7159	6	6	0.6628
whisper_v3	voter	2611	2.6018	1.3488	2.0014	1.1836	7	7	0.6004

step3_model_dropout_validated_ensemble_librispeech

model	times_filtered_out	times_in_pool	filtered_out_rate
seamless_m4t	2114	13055	0.1619
sb	1499	13055	0.1148
whisper_v3	1269	13055	0.0972
parakeet	824	13055	0.0631
cohere	669	13055	0.0512

step3_runs_validated_ensemble_librispeech

run	n_voters	n_samples	filter_fire_rate	tie_rate	refsel_longest_match_validation	refsel_most_common_words
full_pool	5	2611	0.4025	0.0031	0.936	0.064
LOO_minus_seamless_m4t	4	2611	0.2827	0.004	0.9272	0.0728
LOO_minus_whisper_v3	4	2611	0.3095	0.0047	0.9207	0.0793
LOO_minus_parakeet	4	2611	0.3275	0.0046	0.9138	0.0862
LOO_minus_sb	4	2611	0.3052	0.0038	0.9291	0.0709
LOO_minus_cohere	4	2611	0.3336	0.0048	0.918	0.082

step4_pairs_pseudo_validated_ensemble_librispeech

pair	adjacent	gap_wer	preserved_frac	resolved_95
canary < conformer	True	0.0688	0.8394	False
cohere < parakeet	True	0.0741	0.9308	False
parakeet < canary	True	0.0918	0.9434	False

pair	adjacent	gap_wer	preserved_frac	resolved_95
seamless_m4t < whisper_v3	True	0.0949	0.6378	False
parakeet < conformer	False	0.1606	0.9964	True
cohere < canary	False	0.1659	0.9965	True
cohere < conformer	False	0.2346	0.9998	True
sb < seamless_m4t	True	0.275	0.9999	True
conformer < sb	True	0.3136	1	True
sb < whisper_v3	False	0.3699	0.9641	True
canary < sb	False	0.3823	1	True
parakeet < sb	False	0.4741	1	True
cohere < sb	False	0.5482	1	True
conformer < seamless_m4t	False	0.5886	1	True
canary < seamless_m4t	False	0.6574	1	True
conformer < whisper_v3	False	0.6835	1	True
parakeet < seamless_m4t	False	0.7492	1	True
canary < whisper_v3	False	0.7523	1	True
cohere < seamless_m4t	False	0.8232	1	True
parakeet < whisper_v3	False	0.8441	1	True
cohere < whisper_v3	False	0.9182	1	True

step4_pairs_true_validated_ensemble_librispeech

pair	adjacent	gap_wer	preserved_frac	resolved_95
seamless_m4t < whisper_v3	True	0.0325	0.5275	False
canary < conformer	True	0.1414	0.9871	True
conformer < sb	True	0.1738	0.9965	True
parakeet < canary	True	0.2235	1	True
canary < sb	False	0.3152	1	True
sb < seamless_m4t	True	0.3629	1	True
parakeet < conformer	False	0.3649	1	True
sb < whisper_v3	False	0.3954	0.9733	True
cohere < parakeet	True	0.4432	1	True
conformer < seamless_m4t	False	0.5368	1	True
parakeet < sb	False	0.5387	1	True
conformer < whisper_v3	False	0.5693	0.9993	True
cohere < canary	False	0.6667	1	True
canary < seamless_m4t	False	0.6781	1	True
canary < whisper_v3	False	0.7106	1	True
cohere < conformer	False	0.808	1	True
parakeet < seamless_m4t	False	0.9016	1	True

pair	adjacent	gap_wer	preserved_frac	resolved_95
parakeet < whisper_v3	False	0.9341	1	True
cohere < sb	False	0.9819	1	True
cohere < seamless_m4t	False	1.3448	1	True
cohere < whisper_v3	False	1.3773	1	True

step5_baselines_validated_ensemble_librispeech

reference	n_models	spearman	kendall_b	spearman_exact_p_onetailed	kendall_exact_p_onetailed	p_method	n_permutations	min_attainable_p	mean_abs_wer_err
LOO ensemble (proposed)	7	1	1	0.0002	0.0002	exact	5040	0.0002	0.5494
full ensemble (biased)	7	1	1	0.0002	0.0002	exact	5040	0.0002	0.7648
single model: seamless_m4t	6	1	1	0.0014	0.0014	exact	720	0.0014	0.4404
LOO ensemble matched to n-1 (vs seamless_m4t)	6	1	1	0.0014	0.0014	exact	720	0.0014	0.5306
single model: whisper_v3	6	1	1	0.0014	0.0014	exact	720	0.0014	0.5616
LOO ensemble matched to n-1 (vs whisper_v3)	6	1	1	0.0014	0.0014	exact	720	0.0014	0.541
single model: parakeet	6	0.9429	0.8667	0.0083	0.0083	exact	720	0.0014	0.3735
LOO ensemble matched to n-1 (vs parakeet)	6	1	1	0.0014	0.0014	exact	720	0.0014	0.556
single model: sb	6	0.9429	0.8667	0.0083	0.0083	exact	720	0.0014	0.2294
LOO ensemble matched to n-1 (vs sb)	6	1	1	0.0014	0.0014	exact	720	0.0014	0.5452
single model: cohere	6	1	1	0.0014	0.0014	exact	720	0.0014	0.4138
LOO ensemble matched to n-1 (vs cohere)	6	1	1	0.0014	0.0014	exact	720	0.0014	0.6175

step6_contribution_validated_ensemble_librispeech

model_removed	delta_ensemble_wer	delta_ranking_spearman	helps_reference_quality	helps_ranking	delta_pairs_wrong	delta_pairs_wrong_resolvable
seamless_m4t	-0.0516	0	False	True	0	0
whisper_v3	0.0038	-0.009	True	False	1	1
parakeet	0.0745	-0.0714	True	False	2	2
sb	0.0325	0	True	True	0	0
cohere	0.1834	0	True	True	0	0

step6_pools_validated_ensemble_librispeech

pool	k	ensemble_wer_vs_true	ensemble_cer_vs_true	ranking_spearman	ranking_exact_p	ranking_kendall_b	ranking_kendall_exact_p	n_pairs_wrong	n_pairs_wrong_resolvable	worst_wrong_gap_wer	worst_wrong_pair
parakeet+sb+cohere	3	1.3162	0.4072								
whisper_v3+parakeet+sb+cohere	4	1.3181	0.422	1	0.0002	1	0.0002	0	0	0	
whisper_v3+parakeet+cohere	3	1.32	0.3968								
whisper_v3+sb+cohere	3	1.3219	0.4047								
seamless_m4t+whisper_v3+parakeet+sb+cohere	5	1.3697	0.4155	1	0.0002	1	0.0002	0	0	0	

pool	k	ensemble_wer_vs_tr ue	ensemble_cer_vs_tr ue	ranking_spear man	ranking_exact _p	ranking_kendall_ b	ranking_kendall_exact_ p	n_pairs_wro ng	n_pairs_wrong_resolvabl e	worst_wrong_gap_ wer	worst_wrong_p air
seamless_m4t+parakeet+sb+cohere	4	1.3735	0.4372	0.991	0.0004	0.9759	0.0004	1	1	0.1414	canary conforme r
seamless_m4t+parakeet+cohere	3	1.4002	0.4329								
seamless_m4t+whisper_v3+parakeet+cohere	4	1.4021	0.4372	1	0.0002	1	0.0002	0	0	0	
seamless_m4t+whisper_v3+sb+cohere	4	1.4442	0.4581	0.9286	0.0034	0.8095	0.0054	2	2	0.4432	cohere parakeet
whisper_v3+parakeet+sb	3	1.4575	0.4574								
seamless_m4t+sb+cohere	3	1.4938	0.4614								
seamless_m4t+whisper_v3+cohere	3	1.5263	0.4672								
seamless_m4t+whisper_v3+parakeet+sb	4	1.553	0.4838	1	0.0002	1	0.0002	0	0	0	
seamless_m4t+whisper_v3+parakeet	3	1.6161	0.5116								
seamless_m4t+parakeet+sb	3	1.6294	0.5069								
seamless_m4t+whisper_v3+sb	3	1.7211	0.5484								

step6_wrong_pairs_validated_ensemble_librispeech

pool	k	pair	true_winner	pred_winner	wer_true_a	wer_true_b	wer_pseudo_a	wer_pseudo_b	gap_wer_true	gap_wer_pseudo	human_resolvable
seamless_m4t+whisper_v3+sb+cohere	4	cohere parakeet	cohere	parakeet	1.2245	1.6677	1.2377	1.1573	0.4432	0.0804	True
seamless_m4t+whisper_v3+sb+cohere	4	canary conformer	canary	conformer	1.8912	2.0325	1.4133	1.4075	0.1414	0.0057	True
seamless_m4t+parakeet+sb+cohere	4	canary conformer	canary	conformer	1.8912	2.0325	1.3196	1.3196	0.1414	0	True

step7_floor_validated_ensemble_librispeech

model	frac_zero_wer_samples	kendall_b_all	kendall_b_nonzero	n_nonzero	kendall_b_editdist
cohere	0.8407	0.4076	0.2388	416	0.4175
parakeet	0.792	0.5739	0.3158	543	0.5928
canary	0.7641	0.6438	0.3266	616	0.6631
conformer	0.7472	0.6849	0.3547	660	0.7031
whisper_v3	0.7465	0.6433	0.3579	662	0.6674
sb	0.7442	0.653	0.4146	668	0.6751
seamless_m4t	0.6967	0.744	0.4939	792	0.7603

step8_permutation_validated_ensemble_librispeech

perm	order	n	identical	differing	identical_frac
0	cohere+sb+parakeet+whisper_v3+seamless_m4t	2611	2611	0	1
1	parakeet+cohere+sb+seamless_m4t+whisper_v3	2611	2611	0	1
2	seamless_m4t+cohere+parakeet+sb+whisper_v3	2611	2611	0	1
3	cohere+whisper_v3+sb+parakeet+seamless_m4t	2611	2611	0	1
4	cohere+seamless_m4t+parakeet+sb+whisper_v3	2611	2611	0	1

perm	order	n	identical	differing	identical_frac
5	whisper_v3+sb+parakeet+seamless_m4t+cohere	2611	2611	0	1